Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/28/2025
PI Name:	Moudy, Sarah Ph.D.		
Project Title:	Development of Sensorimotor Fitness for Duty Assessments Using Ground Analogs		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sarah.c.moudy@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	
Organization Name:	KBR Wyle/NASA Johnson Space Center		
PI Address 1:	2400 NASA Parkway		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	05/27/2022	**End Date:**	09/30/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	This investigation is a continuation of a Directed Research project with the same name under PI Marissa Rosenberg. Dr. Rosenberg has left NASA. The new PI is Sarah C. Moudy, Ph.D. with KBR/Aegis Aerospace. Millard Reschke, Ph.D. has retired from NASA so he no longer serves as Co-Investigator.		
COI Name (Institution):	Wood, Scott Ph.D. (NASA Johnson Space Center) Peters, Brian Ph.D. (NASA Johnson Space Center) Clark, Torin Ph.D. (University of Colorado, Boulder) Schubert, Michael (Johns Hopkins University)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The first aim will develop a Sensorimotor Disorientation Analog (SDA) that can provide different levels of acute disorientation through combined vestibular, visual, and proprioceptive disruptions. The SDA will be used to increase the range of performance, simulating the moderate-to-severe performance decrements observed postflight. The levels of SDA will be titrated and validated by 1) using feedback from experienced astronauts, and 2) comparison to gold standard measures that have a wealth of spaceflight data at different time points during recovery. Astronaut postflight experience will provide a more realistic starting point for validation of the SDA. Referencing existing postflight data from the gold standard measures will also help us characterize how each magnitude of SDA compares to recovery from long-term microgravity exposure. The second aim will assess the suitability of a proposed set of sensorimotor assessment tasks, or measures that would be feasible within the limited time, resources, and space of a lunar/Martian lander. The assessment tasks should aid in progressive adaptation to the novel gravitational environment, provide opportunities to develop strategies to recover from off-nominal body positions, and mimic operational tasks such that crew can self-assess their potential ability to complete their missions. We will obtain performance of the sensorimotor assessment tasks at varying levels of the SDA magnitude to map sensorimotor ability to the probability of completion of operational performance measures. Our third aim involves up to 90 minutes of prolonged +3GX centrifugation to mimic the vestibular alterations observed after gravity transitions. Immediately after egress from the centrifuge, subjects will perform a subset of the sensorimotor assessment tasks followed by the operational performance tasks. We will map performance in sensorimotor assessment tasks to performance in operational performance measures, similar to aim two. Our fourth aim will utilize the NASA Johnson Space Center (JSC) Active Response Gravity Offload System (ARGOS) to characterize the effects of a reduced gravity load on balance-related exploration and operational measures. The fifth and final aim will build on the lunar landing simulations that will be developed for the Manual Crew Override study (PI: Wood) and the human lander systems training simulations to support the Flight Operations Directorate for exploration class mission training. The study will involve two pre-test familiarization sessions focused on developing proficiency to perform the landing task through multiple landings with both the full simulator and a Just In Time (JIT) trainer platform. Following a standard duration mission timeline (~6-8 months), subjects will perform the same lunar landing while experiencing sensorimotor disorientation (galvanic vestibular stimulation). Between the familiarization and final test sessions, subjects will undergo 1 of 3 “inflight” training paradigms using the JIT trainer including: 1) no training, 2) consistent training throughout the mission, or 3) training only at the end of mission consistent with current ISS protocols. The outcome of this aim is to characterize the relationship between the performance on the JIT platform with post-disorientation lunar landing performance to inform what performance threshold is required to mitigate risk associated with vertigo and disorientation with manual crew override during landings. The main deliverables from this project will be recommended sensorimotor assessments for extravehicular activity (EVA) operations that provide a quantitative index of readiness to perform key operational tasks and validation of lunar landing simulations for training manual crew override during landings. NOTE: Per the Principal Investigator (PI): During the first project year, Aim 1c was removed from the project (Ed., 4/18/23). Aim 1c was to examine a large number of proposed sensorimotor assessment tasks in an effort to reduce the set of tasks to those that are the most sensitive to changes in sensorimotor disruption via the SDA.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This project is in direct response to the baselined Human Research Program (HRP) Path to Risk Reduction milestone of providing updates to the NASA Fitness For Duty Standards. The new standards should be tied to fitness for duty for exploration tasks and provide a quantitative index of readiness to perform key exploration tasks. This research effort will leverage expertise based upon HRP-funded flight research investigations including Functional Tasks Test, Field Test, Standard Measures, and Manual Control, as well as MedB computerized dynamic posturography. This project will leverage critical mission tasks previously established by Ryder et al. ("A novel approach for establishing fitness standards for occupational task performance." Eur J Appl Physiol, 2019) for standards related to the risk of reduced muscle mass, strength, and endurance. This project must also leverage experience with vestibular spaceflight analogs (e.g., Galvanic vestibular stimulation, sustained 3Gx centrifugation) to characterize how the deconditioned state following G-transitions (e.g., postural instability, motion sickness, head movement restrictions) map to functional performance.
Research Impact/Earth Benefits:	This research addresses the Risk of Altered Vestibular/Sensorimotor Function Impacting Critical Mission Tasks (Sensorimotor Risk) and associated research gaps from NASA’s Human Research Roadmap. The development of sensorimotor assessments for EVA and manual control addresses each gap associated with the characterization of the effects of spaceflight on the sensorimotor and neuro-vestibular systems, including postural control and locomotion (Gap SM-101), manual control (Gap SM-102), spatial orientation and motion sickness (Gap SM-103), and other brain functions such as sleep, cognition, attention (Gap SM-104). This project is in direct response to the baselined Human Research Program (HRP) Path to Risk Reduction milestone of defining sensorimotor assessments that can provide a quantitative index of readiness to perform key exploration tasks. The Flight Operations Directorate has identified the need for better sensorimotor assessment tools that would protect against hazards of performing operational tasks too soon following gravitational transitions. The project outcomes will address this recommendation by providing a validated set of sensorimotor assessment tasks that will 1) allow for self-assessment when communication with ground support may be limited or delayed and 2) provide progressive adaptation to the novel gravitational environment. For Earth-based benefits: The sensorimotor community has a need for simple assessment tasks that could be used clinically, through telemedicine, or by the patient alone. These tasks could be used to help in self-determination of the patient’s sensorimotor ability to perform daily activities and/or guide recovery of vestibular-impaired patients.

Task Progress:	During this reporting period, Aim 2, Phase 2, and Aim 3 were completed. Results were presented at the Human Research Program (HRP) Investigators' Workshop (IWS) 2025 and the Vestibular Oriented Research meeting 2025. Aims 2 & 3 sought to map performance in the proposed set of sensorimotor assessment tasks with performance during operational analog tasks while experiencing varying levels of sensorimotor disorientation. Aim 2 utilized the validated sensorimotor disorientation analog (SDA) to induce disorientation, while Aim 3 used sustained +3Gx centrifugation. Validation of the proposed set of sensorimotor assessment tasks will ultimately downselect the task set to those that meet the following criteria: 1) have a strong correlation between performance in the sensorimotor assessment tasks with operational analog tasks, and 2) elicit distinct performance groupings across varying levels of sensorimotor disorientation. Aim 2 Methods Aim 2 Phase 1 was completed with 21 subjects in the previous reporting period. An interim analysis suggested that additional subjects were required to achieve an appropriate statistical power for analyses. Aim 2 Phase 2 was completed on an additional 20 subjects, for a total of 41 subjects. In brief, subjects attended two sessions separated by a minimum of 1 day. The first session provided familiarization to the sensorimotor assessment tasks and operational analog tasks and then baseline data were collected. The second session completed the assessment and operational tasks under two levels of the SDA: 1) high-level replicating crewmember immediate postflight performance (R+0) and 2) low-level replicating R+1 when crewmembers are within the recovery phase yet still have slight sensorimotor disorientation. Aim 3 Methods Data collection occurred at the KBR Aerospace Environment Protection Lab (AEPL) located in San Antonio, TX. The KBR AEPL centrifuge has a free-swinging gondola at the end of the long arm, resulting in a constant alignment of the gravito-inertial force with the gondola. The subjects were oriented in a supine position in the gondola (feet pointing in the direction of motion) so the resultant force vector was in the naso-occipital direction to minimize orthostatic stress. Moderate 0.1 G/s acceleration and deceleration rate were used. Subjects attended two sessions separated by a minimum of 1 day. A familiarization session was conducted including a 5-min centrifuge exposure to minimize any biases toward prior exposure. The centrifuge profile used was unique to those with prior exposure; therefore, no bias or skew in the data was present. The test session included a pre-centrifuge baseline data collection, followed by the 60-min centrifugation spin, and finally, two post-centrifuge tests separated by ~45 min, mimicking the two postflight time points and SDA levels (R+0/high, R+1/low) from Aims 1 & 2. The second post-centrifuge recovery test performance was not significantly different from pre-centrifuge performance; therefore, the following preliminary results will only include the immediate post-centrifuge data. Aim 2 & 3 Preliminary Results To aid in down selection of sensorimotor assessment tasks, we first conducted paired t-tests to examine which tasks were most sensitive to change in sensorimotor disruption. For the SDA, significant differences were present for all sensorimotor assessment and operational analogs tasks between the none (baseline) and high level. There were a few significant differences between the low level and either the none or high level. This does indicate that the low level is able to elicit performance differences in between none and high levels; however, it is not consistent across tasks likely due to subject variability in performance and susceptibility to the SDA. For the centrifuge, only four assessment tasks were significantly different between pre- and immediate post-centrifuge testing. These tasks were consistent with the SDA tasks that had significant performance differences across none, low, and high levels. There were no significant differences in performance for the operational analog tasks post-centrifuge. Given that the operational tasks were conducted after the assessment tasks, the impact of +3Gx on the operational tasks was limited due to the quicker-than-expected recovery. To examine the predictive ability of the sensorimotor assessment tasks to performance in operational analog tasks, preliminary analyses were conducted using the Aim 2 SDA dataset. Pearsons' correlation r-values were calculated between 1) all assessment tasks and 2) each assessment and operational analog task, across individuals and SDA levels (n = 120). Strong correlations were present between each assessment task ($r \geq 0.59$). The two assessment tasks with the strongest correlations to both the operational analog tasks were the fall recovery ($r \geq 0.45$) and obstacle turn tasks ($r \geq 0.60$). Leave-one-out cross-validation (LOOCV) was conducted with a linear model of assessment tasks selected using stepwise regression. The operational analog predictors were the time to complete the capsule egress and the number of bag transfers completed during the 5-minute EVA traversal. For capsule egress, only the obstacle turn task was included in the model with a mean absolute error (MAE) of 12.8 sec between predicted and observed values ($Q2 = 0.35$). For the EVA traversal, the fall recovery, obstacle turn, and four-square-step-test were included in the model with an MAE of 6.5 bags ($Q2 = 0.53$). These results provide early evidence for the down-selection of assessment tasks that were best able to predict performance in operationally relevant tasks for use in upcoming Artemis missions.
Bibliography Type:	Description: (Last Updated: 04/01/2026)
Abstracts for Journals and Proceedings	Moudy SC, Peters BT, Clark TK, Schubert MC, Weiss HM, Wood SJ. "Development and validation of sensorimotor assessment tasks using spaceflight analogs." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Moudy SC, Clark TK, Peters BT, Schubert MC, Wood SJ. "Predicting spaceflight operational performance from sensorimotor assessment tasks." 2025 Vestibular Oriented Research Meeting. Boulder, Colorado, May 19-22, 2025. Abstracts. 2025 Vestibular Oriented Research Meeting. Boulder, Colorado, May 19-22, 2025. , May-2025